WEEK 6

Network Protocols & Communication Models Investigation

Practical Study Report

Field	Details
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Domain	Cybersecurity Fundamentals
Report Type	Practical Investigation & Model Mapping
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This report records observed results from command-line tests and connects those activities to the OSI and TCP/IP communication models. Findings are limited to the evidence collected during this practical.

1. Objective

The objective was to investigate basic network communication using Windows command-line tools, observe DNS resolution and HTTPS request behavior, and map the practical activities to the OSI and TCP/IP models.

2. System and Practical Context

The tests were performed on a Windows computer using nslookup and curl. The target domain was google.com / www.google.com. The results below reflect the command outputs observed during this session only.

3. Tools and Commands Used

Tool / Command	Purpose
nslookup google.com	Queries DNS to obtain IP addresses associated with a domain name.
curl -I https://www.google.com	Sends an HTTP HEAD request over HTTPS and displays response headers.
curl -v -o NUL https://www.google.com	Displays detailed connection and request steps; directs the response body to NUL.
curl -I --max-time 20 https://www.google.com	Repeats the HEAD request with a maximum time limit of 20 seconds.

4. DNS Lookup Investigation

Command: nslookup google.com. The lookup returned a server address of [local DNS server address redacted], displayed as “Server: UnKnown”, followed by multiple IPv4 addresses and IPv6-format addresses using the 64:ff9b::/96 prefix.

Observation	Interpretation
Multiple addresses were returned for google.com.	The domain-name lookup succeeded and provided addresses the client could attempt to use.
DNS server name displayed as “UnKnown”.	The readable server name was not displayed. This does not mean the query failed.
Some IPv6-format addresses used 64:ff9b::/96.	This prefix can be used for IPv4/IPv6 translation. Its appearance alone is not evidence of an error or attack.

Finding: DNS resolution worked for this test. The output does not reveal the full path or every DNS mechanism used by the network.

5. HTTPS Response Header Investigation

Command: curl -I https://www.google.com. The server returned HTTP/1.1 200 OK with response headers, including fields such as Content-Type, Server, X-Frame-Options, Transfer-Encoding and Alt-Svc.

Interpretation: the HEAD request completed successfully and returned a successful HTTP status. Response headers provide metadata about the response, not the full page content. A 200 status for this request does not guarantee that every other request will succeed.

Evidence handling: the original output included Set-Cookie header values. Cookie values are not reproduced in this report. Before publishing screenshots or raw output publicly, redact cookie values and other session-like tokens.

6. Verbose Connection Investigation

Command: curl -v -o NUL https://www.google.com.

Observed step	Result / meaning
Host resolution	The host resolved to multiple IPv6-format and IPv4 addresses.
Initial connection attempts	Some attempts timed out before a later IPv4 connection succeeded.
Connection to port 443	A connection was established to 142.251.150.119 on port 443.
Protocol negotiation	curl reported that HTTP/1.1 was accepted through ALPN.
Request transmission	The HTTP GET request was sent.
Response reception	The connection was reset while receiving the response; curl reported error 56.

Finding: the verbose GET request did not complete successfully even though a connection was established and the request was sent. The collected output does not identify the cause of the reset.

7. HTTPS Retest

Command: curl -I --max-time 20 https://www.google.com. This HEAD request returned HTTP/1.1 200 OK and completed successfully.

The retest was a HEAD request, while the earlier verbose test sent a GET request; therefore, the two tests were not identical. The successful retest does not establish the cause of the earlier failure.

8. OSI and TCP/IP Model Mapping

Practical activity	Main function observed	OSI layer(s)	TCP/IP layer(s)
nslookup google.com	Domain-name resolution	Application	Application
curl HTTPS request	Web request and response headers	Application; lower layers support delivery	Application; lower layers support delivery
Connection to port 443	Transport endpoint used for HTTPS traffic	Transport	Transport
IP address destination	Addressing and forwarding toward destination	Network	Internet

Important limitation: these command-line tests do not directly capture every OSI layer. This is a conceptual mapping of the main functions involved, not a packet-by-packet measurement of every layer.

9. Practical Findings

• The DNS query returned addresses for the requested domain.

• The first HTTPS HEAD request returned HTTP 200 OK.

• The verbose GET request connected to an IPv4 destination on port 443 and sent the request, but the connection was reset during response reception.

• The later HTTPS HEAD retest returned HTTP 200 OK.

• The available evidence is insufficient to determine why the verbose GET request was reset.

10. Cybersecurity Relevance

A security analyst needs to distinguish successful name resolution, connection establishment, request transmission and response reception. These are separate stages and may fail for different reasons. Command output can support an investigation, but a single timeout or reset is not proof of malware, compromise or a particular network fault.

11. Investigation Summary

Test	Observed status	Conclusion
DNS lookup	Addresses returned	Resolution successful for this test.
HTTPS HEAD request	HTTP 200 OK	Request completed successfully.
Verbose HTTPS GET	Connection reset during receive	Request did not complete successfully; cause unknown.
HTTPS HEAD retest	HTTP 200 OK	Retest completed successfully.

12. Conclusion

This practical linked DNS resolution and HTTPS requests with the OSI and TCP/IP communication models. The observed results included successful DNS resolution and successful HEAD requests, alongside one verbose GET request that was reset during response reception. The report documents what was observed without assuming a cause that the evidence cannot establish.

13. Evidence and Study-Use Note

Evidence collected: screenshots of the DNS lookup, HTTPS response headers, verbose connection test and HTTPS retest, plus the text files containing practical observations and OSI/TCP-IP mapping.

Before placing evidence in a public repository, review screenshots for cookie values, session tokens, private addresses, usernames or other personal information. Redact sensitive values where appropriate. Do not upload unredacted cookie headers.

Prepared for personal cybersecurity learning and portfolio documentation. This educational report is not a formal security audit or a diagnosis of the network.